

Riemann surfaces Problem set 1

1. Let $p \in \mathbb{C}$, let $\mathcal{O}_p$ denote the ring of germs of holomorphic functions at p , let $m_p = \{f \in \mathcal{O}_p \mid f(p) = 0\}$, and let $\mathcal{M}_p$ denote the set of germs of meromorphic functions at p . For $f \in \mathcal{O}_p - \{0\}$, define

$$\text{ord}_p(f) := \min\{n \geq 0 \mid f^{(n)}(p) \neq 0\}.$$

a) Show that $\mathcal{O}_p$ is an integral domain.

b) Show that $\mathcal{M}_p$ is a field, and is the quotient field of $\mathcal{O}_p$.

c) Show that $\mathcal{O}_p^\times = \mathcal{O}_p - m_p$, and hence show that m_p is the unique maximal ideal of $\mathcal{O}_p$.

d) Show that for any $f \in \mathcal{O}_p$, $m_p = (f)$ if and only if $f(p) = 0$ and $f'(p) \neq 0$.

e) Show that for any $f \in \mathcal{O}_p - \{0\}$,

$$\text{ord}_p(f) = \max\{n \geq 0 \mid f \in m_p^n\}.$$

f) Show that for any $f \in \mathcal{O}_p - \{0\}$, $\text{ord}_p(f) = k$ if and only if $f = g^k \cdot u$ for some $g \in \mathcal{O}_p$ such that $m_p = (g)$ and some unit $u \in \mathcal{O}_p^\times$.

g) Let ϕ be a holomorphic function in a neighbourhood of p such that $\phi'(p) \neq 0$, and let $q = \phi(p)$. Show that for any $f \in \mathcal{O}_q - \{0\}$,

$$\text{ord}_p(f \circ \phi) = \text{ord}_q(f).$$

h) Show that for any $f \in \mathcal{O}_p - \{0\}$,

$$\text{ord}_p(f) = \text{Res}(f'/f, p).$$

i) Let $f \in \mathcal{O}_p - \mathbb{C}$, and let ψ be a holomorphic function near $f(p)$ such that $\psi'(f(p)) \neq 0$. Show that

$$\text{ord}_p(\psi \circ f - \psi \circ f(p)) = \text{ord}_p(f - f(p)).$$

2. Let X be a Riemann surface and let $p \in X$. Let $\mathcal{O}_p$ be the ring of germs of holomorphic functions at p and let $m_p = \{f \in \mathcal{O}_p \mid f(p) = 0\}$. For $f \in \mathcal{O}_p - \{0\}$, define $\text{ord}_p(f) := \text{ord}_{\phi(p)}(f \circ \phi^{-1})$, where ϕ is a chart in a neighbourhood of p .

a) Show that $\text{ord}_p(f)$ is well-defined independent of the choice of chart ϕ near p .

b) Show that for $\phi \in \mathcal{O}_p$, $m_p = (\phi)$ if and only if $\phi(p) = 0$ and there are neighbourhoods U, V of $p \in X, 0 \in \mathbb{C}$ respectively such that $\phi|_U$ is a biholomorphic map from U to V .

3. Let X, Y be Riemann surfaces and let $p \in X, q \in Y$. Let f be a nonconstant holomorphic map from a connected neighbourhood of p to a neighbourhood of q such that $f(p) = q$. Define $\deg_p(f) := \text{ord}_{\phi(p)}(\psi \circ f \circ \phi^{-1} - \psi \circ f \circ \phi^{-1}(\phi(p)))$ where ϕ, ψ are charts near p, q respectively.

a) Show that $\deg_p(f)$ is well-defined independent of the choice of charts ϕ, ψ .

b) Show that $\deg_p(f) = k \geq 1$ if and only if there are charts ϕ, ψ near p, q respectively such that $\phi(p) = 0 = \psi(q)$ and $\psi \circ f \circ \phi^{-1}(z) = z^k$ in a neighbourhood of 0.

4. Let $f : X \rightarrow Y$ be a nonconstant holomorphic map between connected Riemann surfaces X, Y .

- a) Show that f is an open mapping.
- b) Show that if X is compact, then Y is compact and f is onto.
- c) Show that any holomorphic function on a compact Riemann surface is constant.
- 5. Show that a holomorphic bijection $f : X \rightarrow Y$ between Riemann surfaces is a biholomorphism.
- 6. Show that any proper holomorphic map $f : \mathbb{C} \rightarrow \mathbb{C}$ is a polynomial.
- 7. Let $P \in \mathbb{C}[z]$ be a polynomial of degree $d \geq 1$. Show that

$$\sum_{x \in P^{-1}(y)} \deg_x(P) = d$$

for all $y \in \mathbb{C}$.

- 8. Let $R = P/Q \in \mathbb{C}(z)$ be a rational function, where P, Q are coprime polynomials such that $d := \max(\deg P, \deg Q) \geq 1$.

- a) Show that

$$\sum_{x \in R^{-1}(y)} \deg_x(R) = d$$

for all $y \in \hat{\mathbb{C}}$.

- b) Show that $R : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ is biholomorphic if and only if R is a Moebius transformation.

- 9. Let X be a compact Riemann surface. Show that for any $f \in \mathcal{M}(X) - \{0\}$, we have

$$\sum_{p \in X} \text{ord}_p(f) = 0.$$

- 10. Let X be a compact Riemann surface of genus $g \geq 1$. Let $p, q \in X$ be distinct points of X and let $D \in \text{Div}_0(X)$ be the degree zero divisor $D = 1 \cdot p - 1 \cdot q$. Show that D is not a principal divisor.

- 11. Show that any degree zero divisor on $\hat{\mathbb{C}}$ is principal.

- 12. Let X_1, X_2 be Riemann surfaces, and let $G_i < \text{Aut}(X_i), i = 1, 2$ be subgroups such that the quotient maps $X_i \rightarrow X_i/G_i, i = 1, 2$ are covering maps.

- a) Let $f : X_1 \rightarrow X_2$ be a biholomorphic map such that $fG_1f^{-1} = G_2$. Show that f induces a biholomorphic map $g : X_1/G_1 \rightarrow X_2/G_2$.

- b) Suppose X_1, X_2 are simply connected. Show that any biholomorphic map $g : X_1/G_1 \rightarrow X_2/G_2$ lifts to a biholomorphic map $f : X_1 \rightarrow X_2$ such that $fG_1f^{-1} = G_2$.

- 13. Let $f, g \in \text{Aut}(\mathbb{H}) - \{id\}$. Show that f, g commute if and only if $\text{Fix}(f) = \text{Fix}(g)$.

- 14. Let $g \in \text{Aut}(\mathbb{H})$ be a hyperbolic element of the form $g(z) = e^R z$ for some $R > 0$. Show that $\mathbb{H}/\langle g \rangle$ is biholomorphic to $A = \{1 < |z| < e^{2\pi^2/R}\}$.

- 15. Use problems 12(b) and 14 above to show that two annuli $A_i = \{1 < |z| < R_i\}, i = 1, 2$ are biholomorphic if and only if $R_1 = R_2$.

- 16. Let $\mathbb{D}^*$ be the punctured unit disk in $\mathbb{C}$.

- a) Show that $\text{Aut}(\mathbb{D}^*) = \{z \mapsto \lambda z \mid \lambda \in S^1\}$.

b) Let $G < \text{Aut}(\mathbb{D}^*)$ be a discrete subgroup. Show that the map $\mathbb{D}^* \rightarrow \mathbb{D}^*/G$ is a covering, and $\mathbb{D}^*/G$ is biholomorphic to $\mathbb{D}^*$.

17. Let A be an annulus $A = \{1 < |z| < R\}$.

a) Show that $\text{Aut}(A)$ is isomorphic to $S^1 \rtimes \mathbb{Z}/2\mathbb{Z}$. (without deck transformation?)

b) Hence show that A is not biholomorphic to $\mathbb{D}^*$.

18. Let X be a connected Riemann surface and suppose that there is a holomorphic covering map $\pi : X \rightarrow \mathbb{D}^*$ of finite degree $n \geq 1$. Show that there is a biholomorphic map $\phi : \mathbb{D}^* \rightarrow X$ such that $\pi \circ \phi(z) = z^n$ for all $z \in \mathbb{D}^*$.